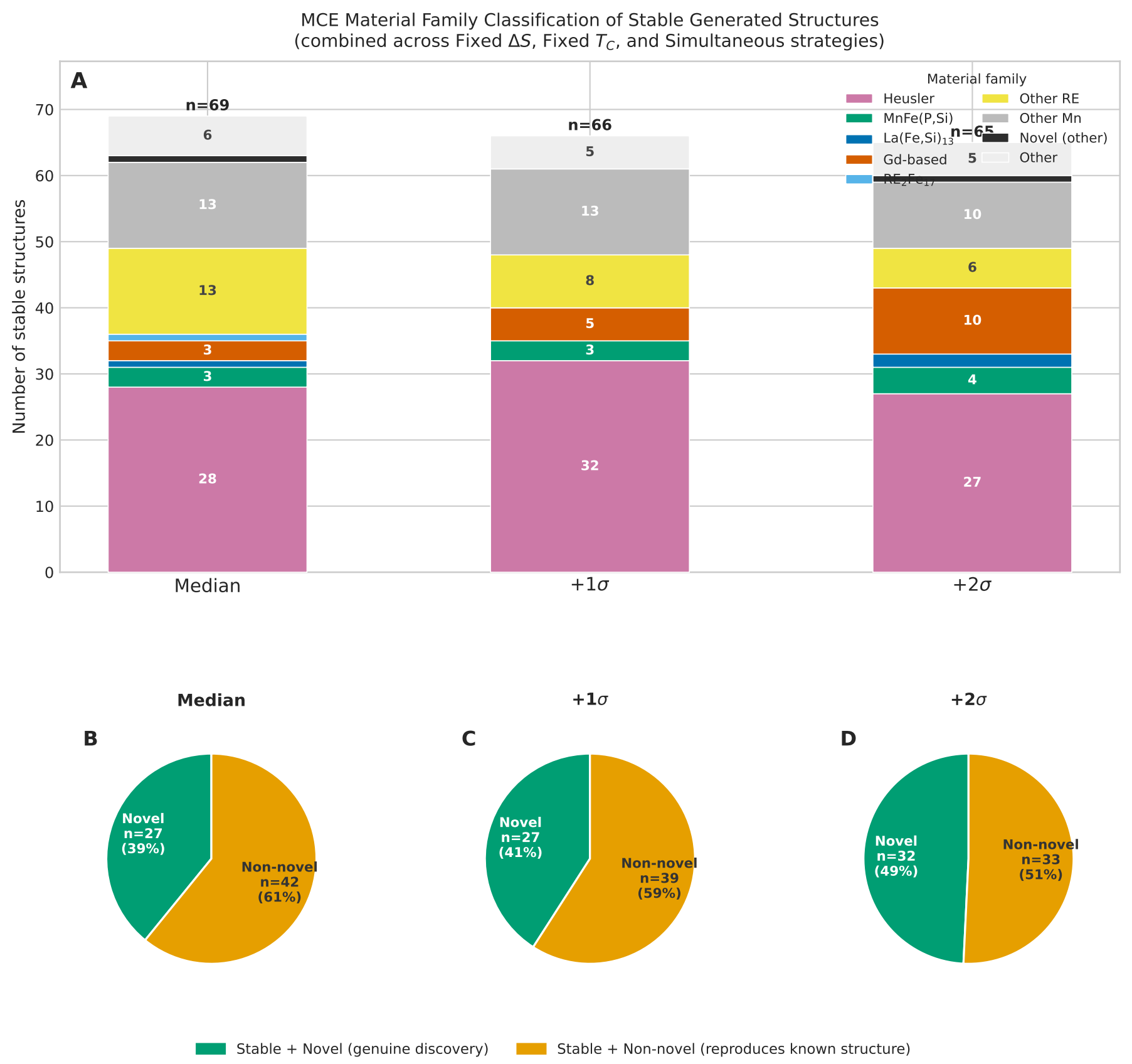
MCE Material Family Classification of Stable Generated Structures
(combined across Fixed ΔS, Fixed $T_C$, and Simultaneous strategies)
A
Material family
Heusler
MnFe(P,Si)
La(Fe,Si)13
Gd-based
RE2Fe17
Other RE
Other Mn
Novel (other)
Other
Number of stable structures
n=69
n=66
n=65
Median
+1σ
+2σ
Median
+1σ
+2σ
B
C
D
Novel n=27 (39%)
Non-novel n=42 (61%)
Novel n=27 (41%)
Non-novel n=39 (59%)
Novel n=32 (49%)
Non-novel n=33 (51%)
Stable + Novel (genuine discovery)
Stable + Non-novel (reproduces known structure)